

Study of Entanglement Growth in Random Quantum Circuits

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CONTENTS



Motivation: Bipartite Entanglement.



Upper Bound: Minimal cut.



Numerical results.



Research In Progress.



Conclusions.

MOTIVATION

Entanglement

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graph TD; A[Entanglement] --> B[How to optimally detect entanglement theoretically and in the laboratory?]; A --> C[How to characterize, control, and quantify entanglement?];
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How to optimally detect entanglement theoretically and in the laboratory?

How to characterize, control, and quantify entanglement?

BIPARTITE ENTANGLEMENT

A bipartite quantum system

$$\hookrightarrow |\psi\rangle = \sum_{i_A, i_B} c_{i_A, i_B} |i_A\rangle \otimes |i_B\rangle \in \mathcal{H} = \mathcal{H}_A \otimes \mathcal{H}_B$$

A bipartite system is entangled
if it cannot be written as

$$\longrightarrow |\psi\rangle \neq |\psi_A\rangle \otimes |\psi_B\rangle$$

And for mixed states: $\left[\rho \neq \rho_A \otimes \rho_B \right]$

ENTROPY AS A METRIC TO MEASURE ENTANGLEMENT

In classical information theory, exist the Shannon entropy.

$$H(X) := - \sum_{x \in \mathcal{X}} p(x) \log p(x)$$



“Number of questions I must ask to find particular information”.

Similarly, for quantum systems the Von Neumann entropy is defined

$$S = -\text{Tr}(\rho \log \rho)$$

Entropy is used to quantify entanglement

$$\left\{ \begin{array}{ll} S = 0 & \text{Separable} \\ S = 1 & \text{Maximally entanglement} \end{array} \right.$$

ENTROPY AS A METRIC TO MEASURE ENTANGLEMENT

📦 Rényi entropy generalizes a set of entropies: Calculate tool

$$S_n = \frac{1}{1-n} \log (\text{Tr}(\rho_A^n)) = \frac{1}{1-n} \log \left(\sum_i \lambda_i^n \right)$$

➡ ρ_A is the reduced matrix of system A, and λ_i are eigenvalues of ρ_A

$$\square \quad n = 0 \quad : \quad S_0 = \log (\text{rank}(\rho_A))$$

$$\square \quad n \rightarrow \infty \quad : \quad S_\infty = \log \left(\frac{1}{\lambda_{\max}} \right)$$

$$\square \quad n \rightarrow 1 \quad : \quad S_1 = S$$

ENTROPY AS A METRIC TO MEASURE ENTANGLEMENT

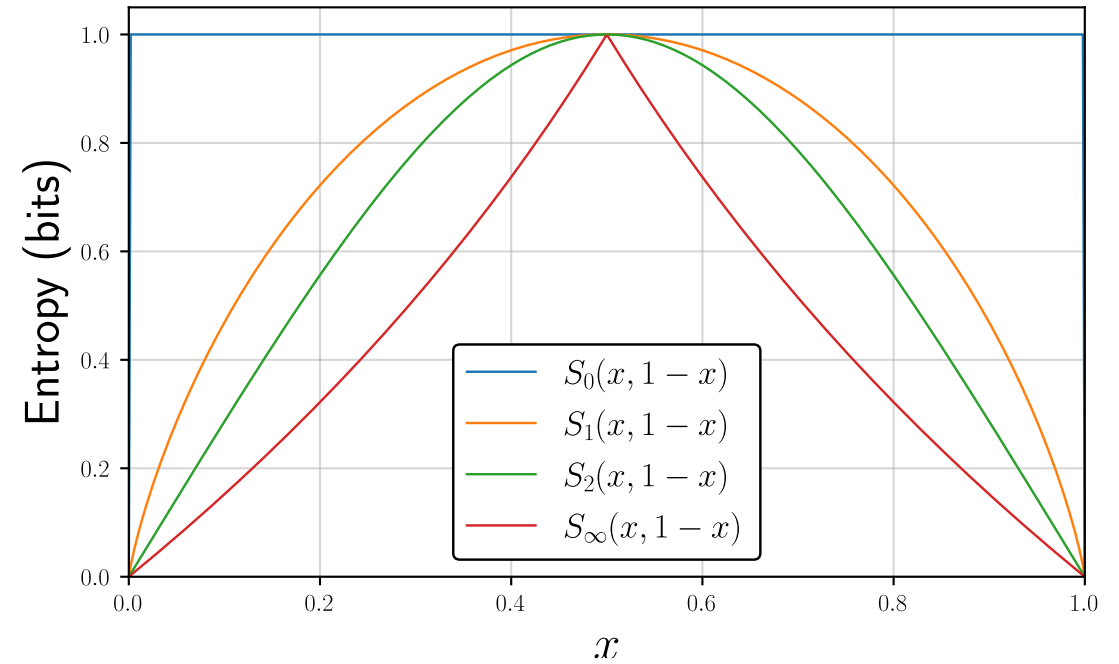
📦 **Rényi entropy** $S_n = \frac{1}{1-n} \log \left(\sum_i \lambda_i^n \right)$

S_n is bounded:

$$0 \leq S_n \leq \log D$$

$$S_{n+1} \leq S_n \leq S_{n-1}$$

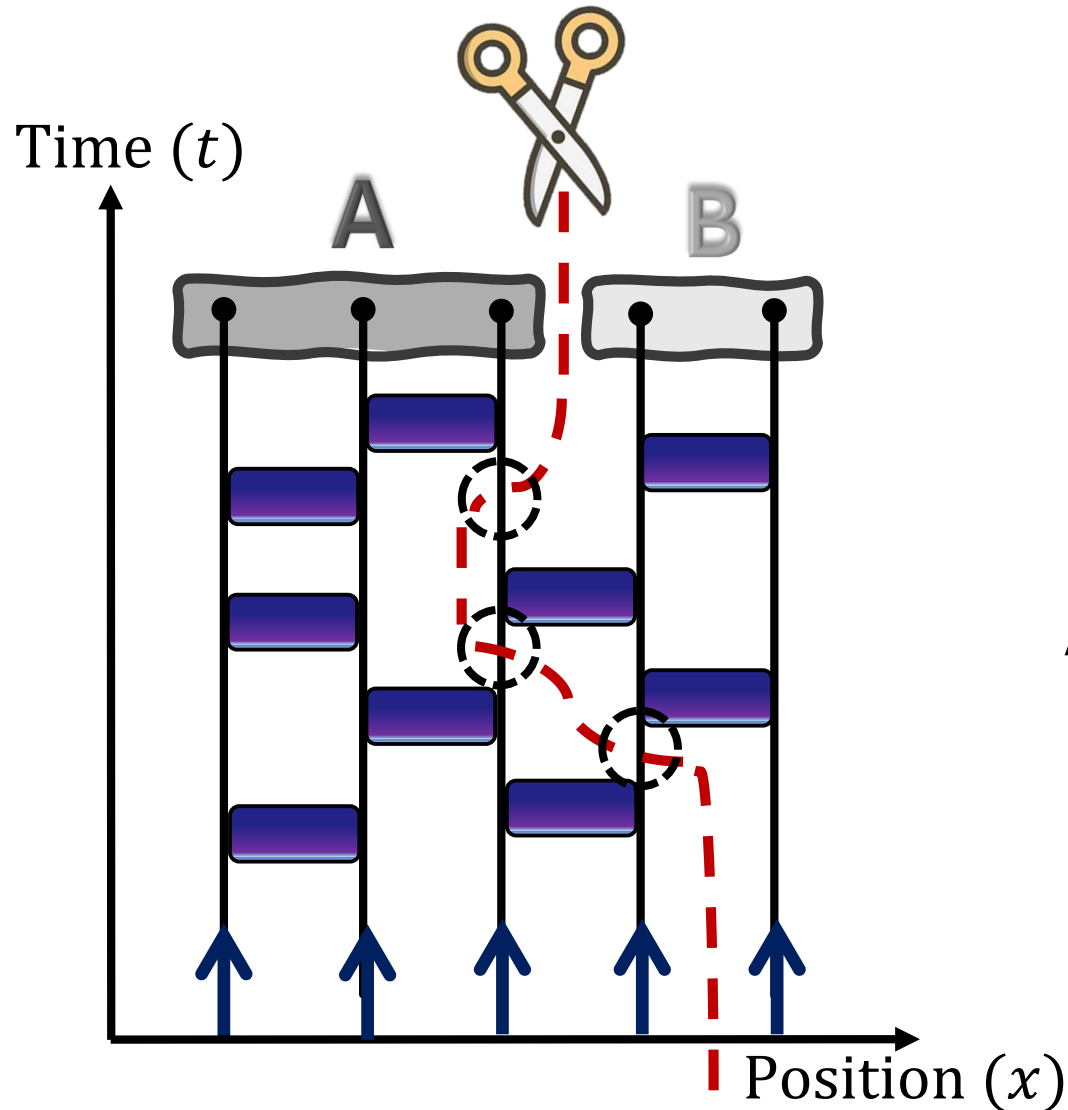
$$S_\infty \leq S_n \leq S_0$$



Rényi entropies for the binary distribution $P=(x, 1-x)$.

$$S_n \leq \log (\# \text{ of nonzero terms in any decomposition})$$

UPPER BOUND: MINIMAL CUT

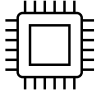


The minimal cut-formalism provides a classical bound to the entanglement growth in **1D-circuits**.

A **classical** upper bound for the bipartite entropies,

$$S_n \leq N_{\text{min cut}}$$

TWO-QUBIT GATES CIRCUIT



When restricted to two-qubit (local) gates, and in the thermodynamic limit, the problem of minimizing the number of cuts is the same as that of finding the energy minima of a polymer in a disordered medium (DPRE).



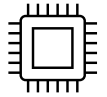
The entanglement entropy across a cut to grow according to the Kardar-Parisi-Zhang (KPZ) equation.

$$\frac{\partial \mathcal{N}}{\partial t} = \nu \partial_x^2 \mathcal{N} - \frac{\lambda}{2} (\partial_x \mathcal{N})^2 + \eta(x, t) + c$$

diffusive smoothing

noise

KPZ MODEL



KPZ scaling is characterized by an exponent β governing the size of fluctuations in the interface height, an exponent α governing spatial correlations.

- The mean entropy (entanglement) grows linearly in time, with a universal subleading correction:

$$\langle \mathcal{N}(x, t) \rangle \sim v_E t$$

- The fluctuations grow as:

$$\Delta \mathcal{N}(x, t) \sim t^\beta$$

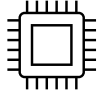
In 1+1 dimension

$$\beta = 1/3, \alpha = 1/2$$

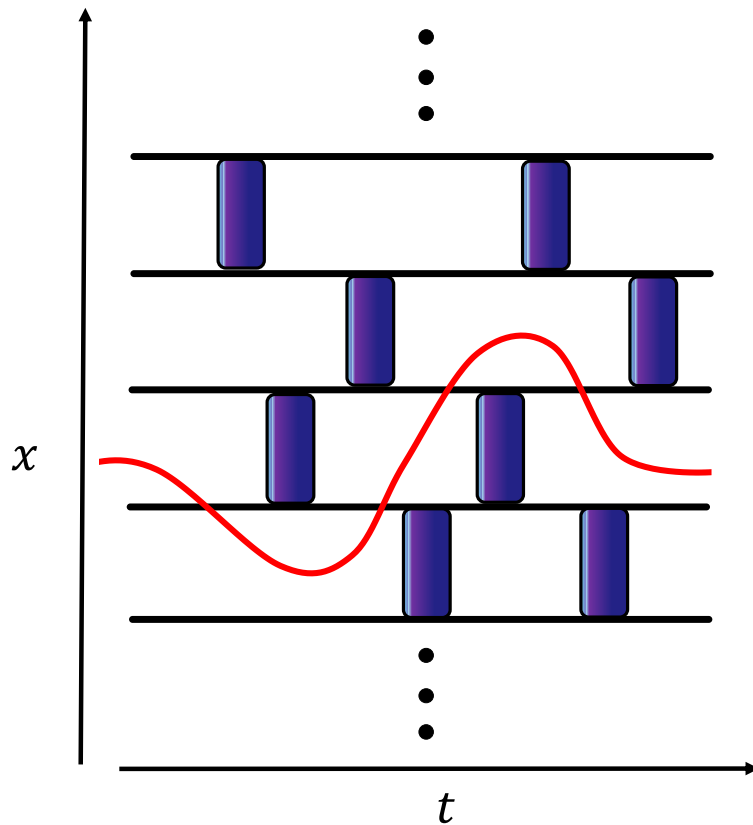
- The correlation length governing spatial correlations in the fluctuations grows with time as:

$$\xi(t) \sim t^{\beta/\alpha}$$

CUT GROWTH MODEL



Any cut, can be view as the word-line traced by a classical point-particle (walker) which avoids the obstacles.



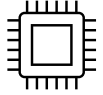
the probability $p(x, t)$ that the “walker” is in the x -th carrel at time t ,

$$\begin{aligned} p(x, t + 1) = & \frac{1}{2} c(x + 1, t) p(x + 1, t) \\ & + \frac{1}{2} c(x - 1, t) p(x - 1, t) \\ & + (1 - c(x, t)) p(x, t) \end{aligned}$$

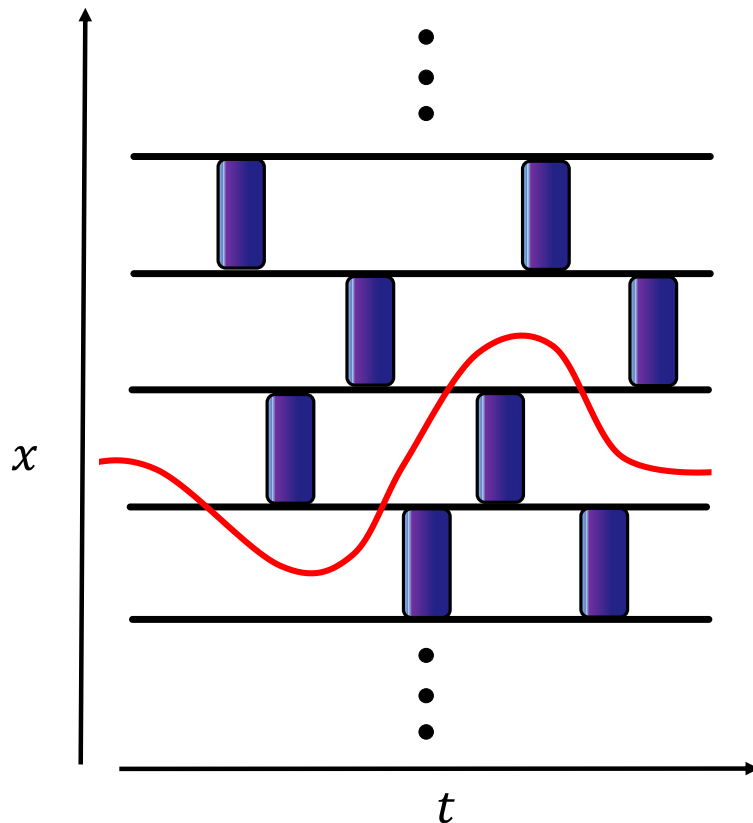
For the case of uniform distributed two-qubit gates,

$$c(x, t) = \frac{1}{L-1}, \text{ with } L \text{ the number of qubits.}$$

CUT GROWTH MODEL



Any cut, can be view as the word-line traced by a classical point-particle (walker) which avoids the obstacles.



$$p(x, t + 1) = \frac{1}{2(L + 1)} (p(x + 1, t) + p(x - 1, t)) + \left(1 - \frac{1}{L + 1}\right) p(x, t)$$

Diffusion constant $D \sim 1/L$, and associated characteristic velocity $v = D/l_{typ}$, being l_{typ} the typical size of the generators (n-qubit gates), which in our case is $l_{typ} = 2$.

CUT GROWTH MODEL

- We are interested in the cut increase distribution $Q(t)$, that is, the number of times the walker changes carrel by avoiding an obstacle:

$$Q(t) = \sum_x c(x, t) p(x, t) = \frac{1}{L-1} \sum_x p(x, t)$$

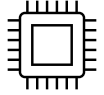
- And we also define $\mathcal{N}_c(t)$ as the cut number distribution in time:

$$\mathcal{N}(t) = \sum_{\tau=0}^t \Delta \mathcal{N}(\tau), \quad Q(t) = \text{Prob}(\Delta \mathcal{N}(t) = 1)$$

- Each cut-gap is an independent-binary random variable with distribution:

$$\varrho(\Delta \mathcal{N}(t)) = Q(t) \delta(\Delta \mathcal{N}(t) - 1) + (1 - Q(t)) \delta(\Delta \mathcal{N}(t))$$

CUT GROWTH MODEL



Therefore, the whole distribution P of the $\mathcal{N}_c(t)$ is given by:

$$P[\mathcal{N}(t)] = \int_{\mathbb{R}} e^{ik\mathcal{N}(t) + \sum_{\tau=0}^t \log(1 + \mathcal{Q}(\tau)[e^{-ik} - 1])} dk$$

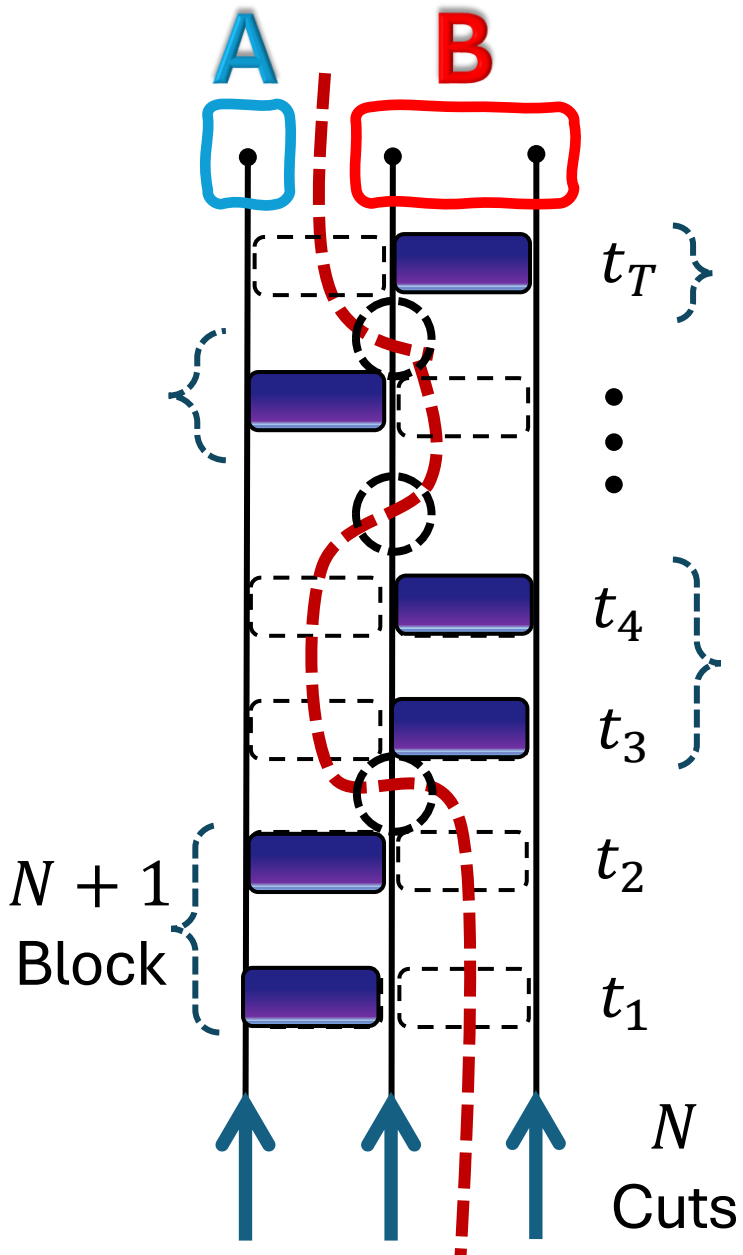
After integrating:

$$P[\mathcal{N}(t)] \propto \frac{1}{\mathcal{N}(t)!} \left(\sum_{\tau=1}^t \frac{\mathcal{Q}(\tau)}{1 - \mathcal{Q}(\tau)} \right)^{\mathcal{N}(t)} \prod_{\tau=1}^t (1 - \mathcal{Q}(\tau))$$

Therefore, the number of cuts $\mathcal{N}$ at deep- t follows the Binomial distribution $\mathcal{N} \sim B(t, Q)$ with $Q = \frac{1}{L+1}$

$$P[\mathcal{N}(t)] = \binom{t}{\mathcal{N}} (1 - Q)^{t-\mathcal{N}} Q^{\mathcal{N}} \longrightarrow \overline{\mathcal{N}(t)} = tQ(t)$$

RANDOM CIRCUITS WITH 3 QUBITS



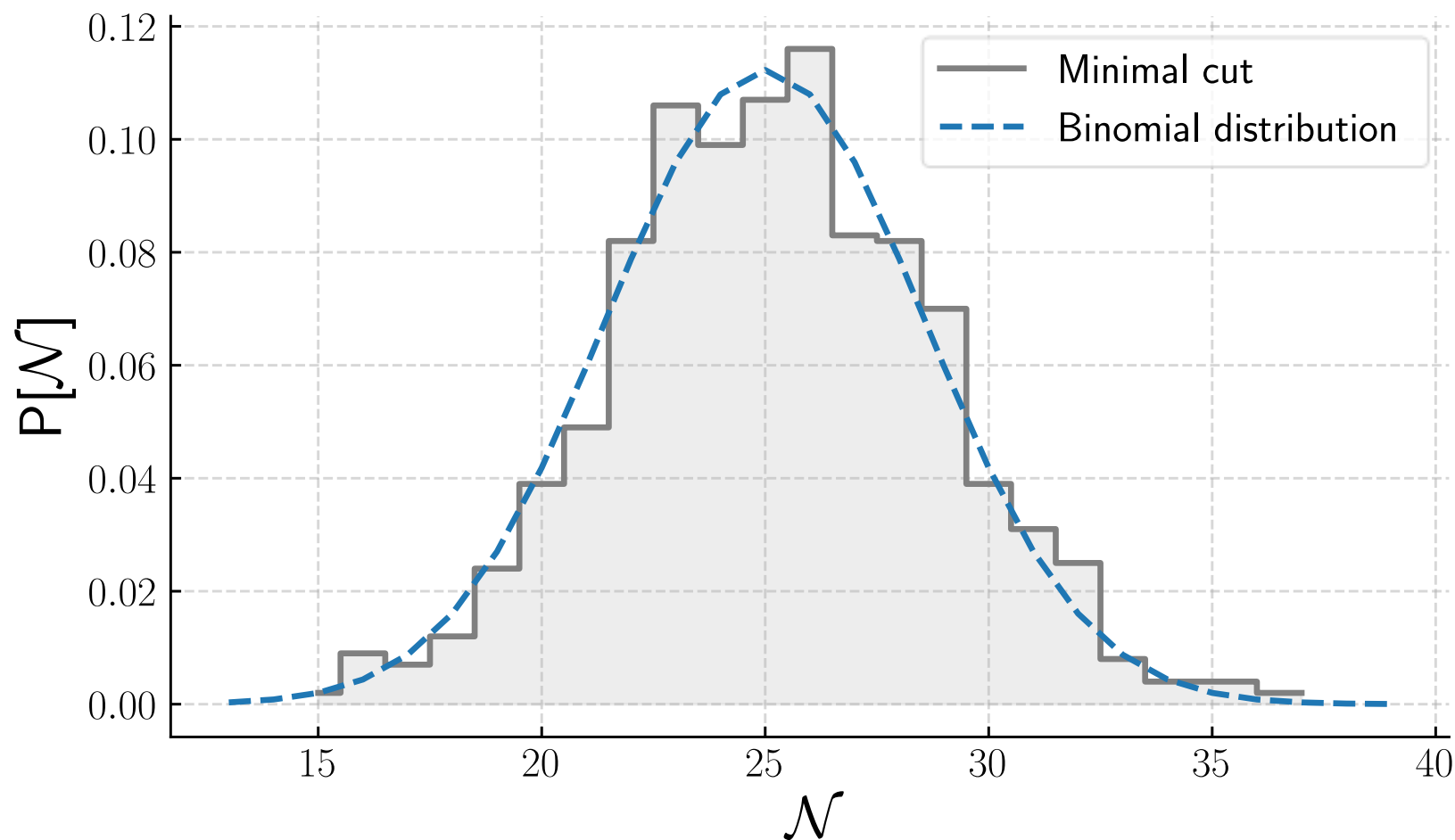
Random circuits with 3 qubits

- There will be a cut for each obstacle (Block) in the circuit.
- Since we have two possible positions, distributing T gates in $N + 1$ blocks becomes a binomial distribution.

$$P_{\text{min cut}}(\mathcal{N}) = \frac{\# \text{ Circuits with } \mathcal{N} + 1 \text{ blocks}}{\# \text{ Total circuits}}$$

$$P_{\text{min cut}}(\mathcal{N}) = \frac{1}{2^L} \binom{T}{\mathcal{N}} = \frac{T!}{2^L \mathcal{N}! (T - \mathcal{N})!}$$

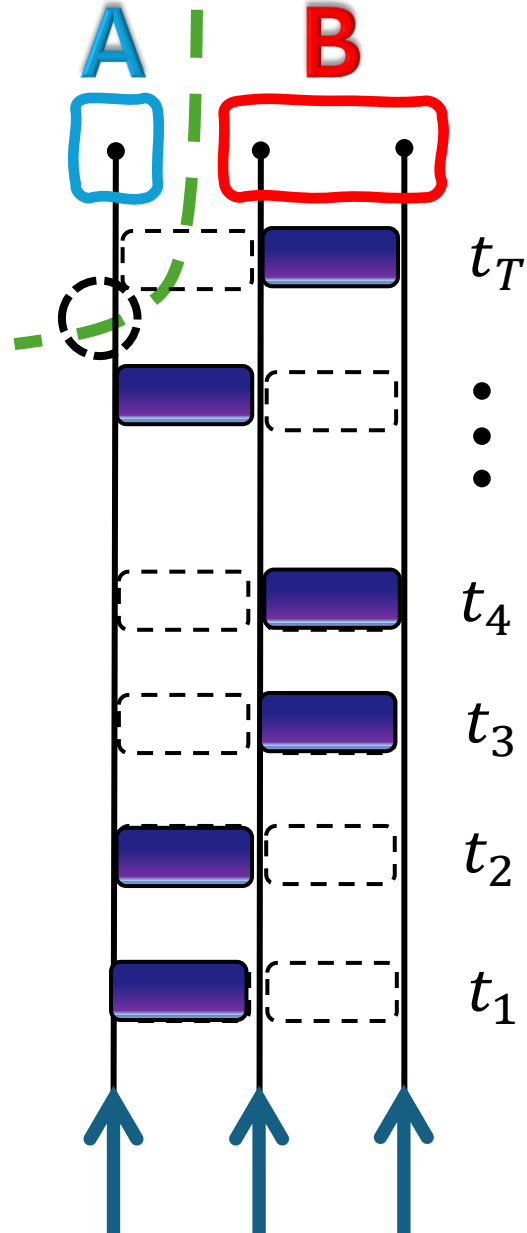
RANDOM CIRCUITS WITH 3 QUBITS



□ obtained from 1000 ensembles of random Clifford circuits without considering saturation effects

The data clearly follows the binomial distribution, with mean value $tQ = \frac{t}{L-1}$

RANDOM CIRCUITS WITH 3 QUBITS

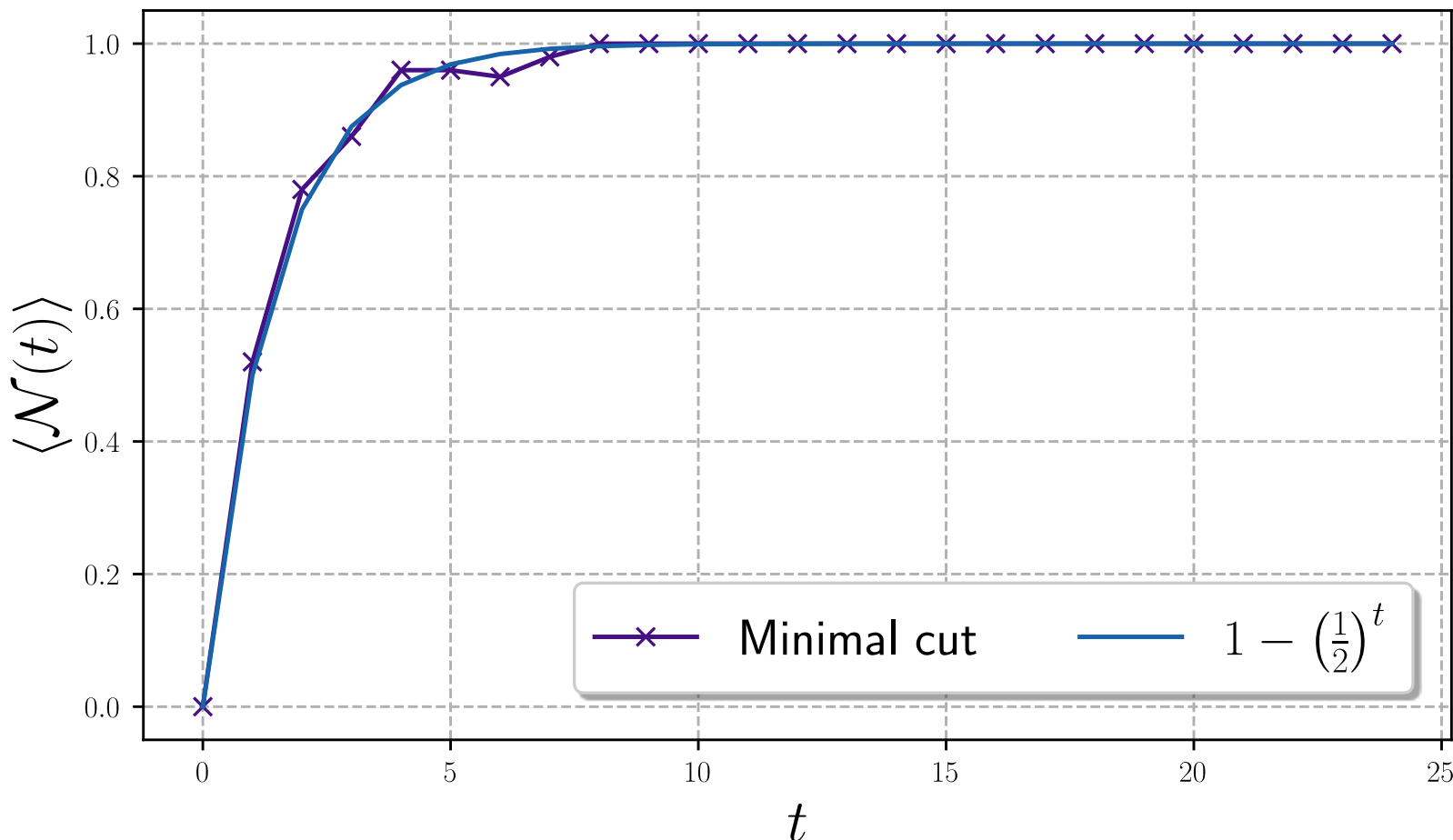


In physical circuits, the minimal cut $\mathcal{N}(x, t)$ cannot grow indefinitely; instead, it saturates at the subsystem size x (with $x \leq L - x$).

Thus, the mean cut is given by the following truncated binomial form:

$$\begin{aligned} \langle \mathcal{N}(x, t) \rangle &= \sum_{n=0}^{x-1} n \binom{t}{n} (1 - \mathcal{Q})^{t-n} \mathcal{Q}^n + x \left(1 - \sum_{n=0}^{x-1} \binom{t}{n} (1 - \mathcal{Q})^{t-n} \mathcal{Q}^n \right) \\ &= \sum_{n=0}^{x-1} (n - x) \binom{t}{n} (1 - \mathcal{Q})^{t-n} \mathcal{Q}^n + x = 1 - \left(\frac{1}{2} \right)^t. \end{aligned}$$

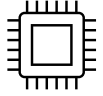
RANDOM CIRCUITS WITH 3 QUBITS



- Average minimal cut for random Clifford circuits with $L = 3$ qubits, computed over 50 ensembles

The data shows clear agreement with the analytical description, once saturation effects are considered.

QUENCHED APPROACH



Both the optimization and averaging are on the same footing, and at the end finds the minima,

$$\min_{P \in P_{\text{all}}} \langle \mathcal{N}[P_c] \rangle_{\mathcal{U}_{L,T}}$$

Where the brackets denote the average over $\mathcal{U}_{L,T}$, the ensemble of all random circuits $c \in \mathcal{U}_{L,T}$ with L qubits and depth T . And $P_c \in P[c]$ the set of all paths compatible with the circuit c . one.



$$\min_{P \in P_{\text{all}}} B(T, \mathcal{Q}) = 0$$



CUT GROWTH MODEL

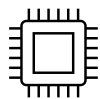
In this model, we only assume that the random walker avoids obstacles when encountering one, which does not deviate significantly from the behavior of a true minimal cut. Hence, we can expect that the **generic** behavior of the **cut** brings us closer to understanding the **minimal** one.

SUMMARY

1. In the pre-saturation regime, the mean cut grows linearly with time, in agreement with the predictions of the KPZ universality class.
2. by employing the binomial approximation for a generic cut, we predict that the entanglement velocity scales inversely with the number of qubits, $v_E \propto 1/L$. This already goes beyond the KPZ framework.

CUT GROWTH MODEL

SUMMARY



Moreover, under the diffusion equation picture introduced earlier, the entanglement velocity must be determined by the gate size $v_E = 1/2L$.

$$S_\alpha(x, t) \leq \mathcal{N}_{\text{minimal}}(x, t) \sim v_E t = \frac{t}{2L}$$

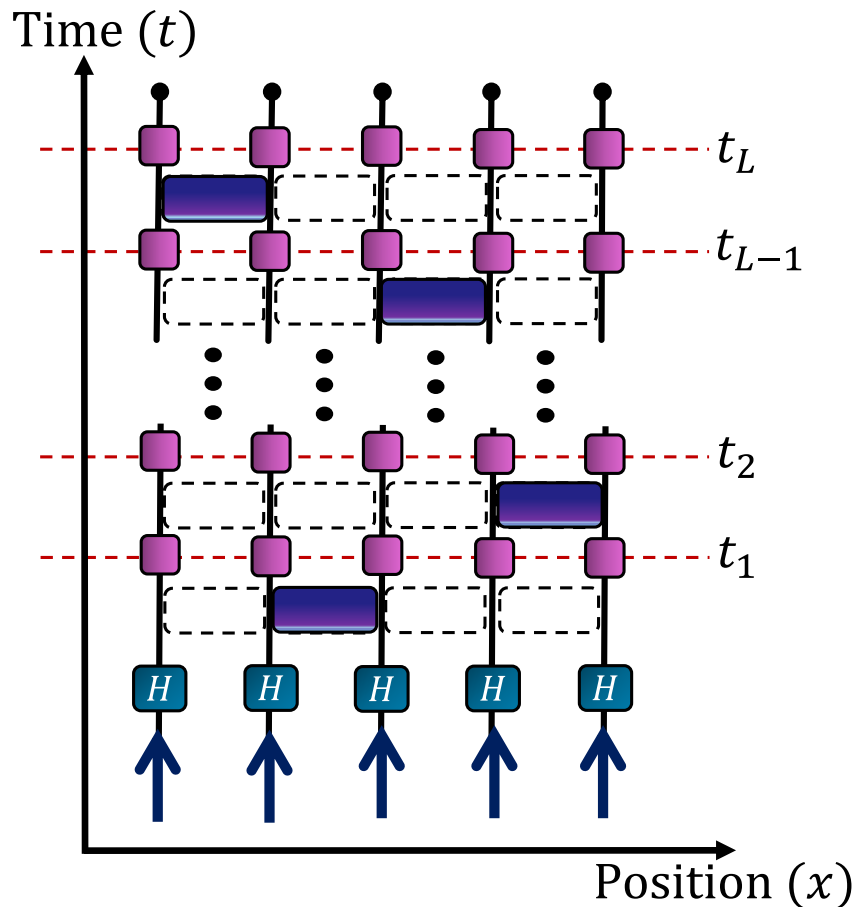


$$\langle \mathcal{N}_{\text{minimal}}(x, t) \rangle \sim \begin{cases} v_e t, & \text{for } x \leq t_{\text{saturation}} \\ x, & \text{for } x > t_{\text{saturation}} \end{cases}$$

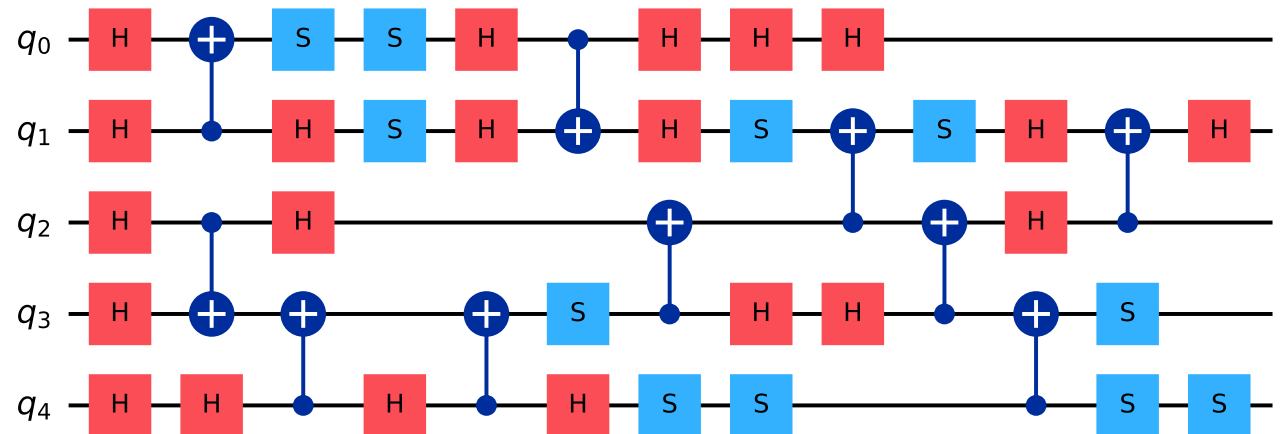


SIMULATION

Clifford circuits, or “stabilizer circuits,” are a special class of quantum circuits that play an important role in quantum information theory.



The Clifford group may be generated by a small set of local Clifford gates: the two-site **controlled NOT gates** and the **single-site Hadamard** and **phase gates**.

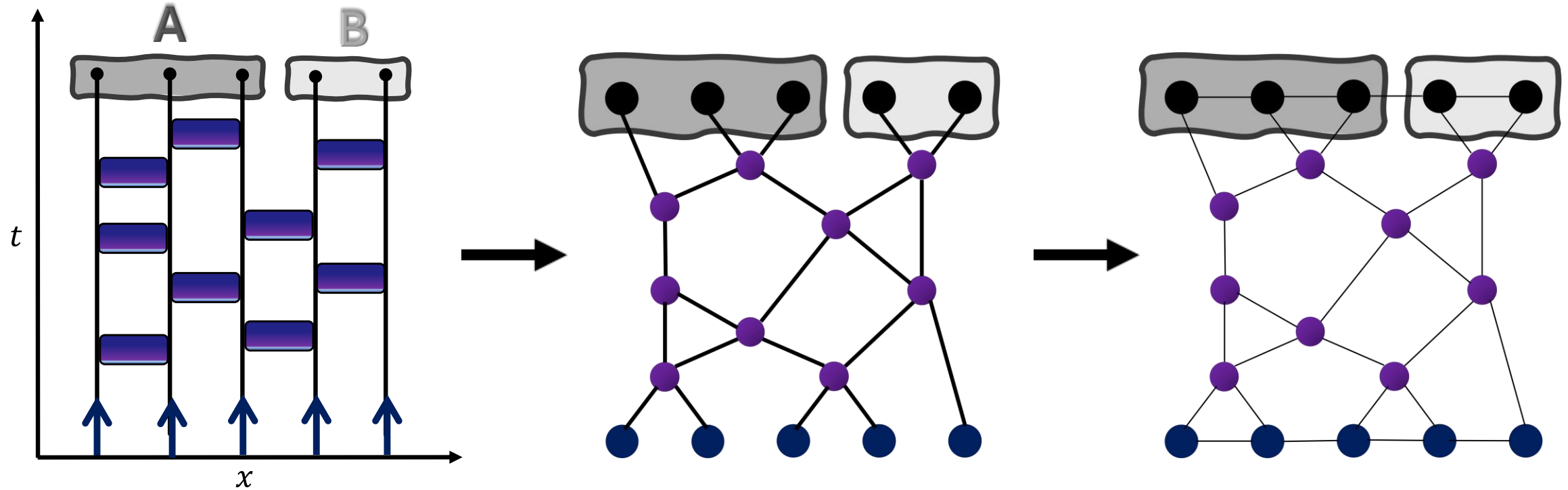


This is an example in Qiskit of a circuit with 5 qubits evolving until $t = 10$.

SIMULATION

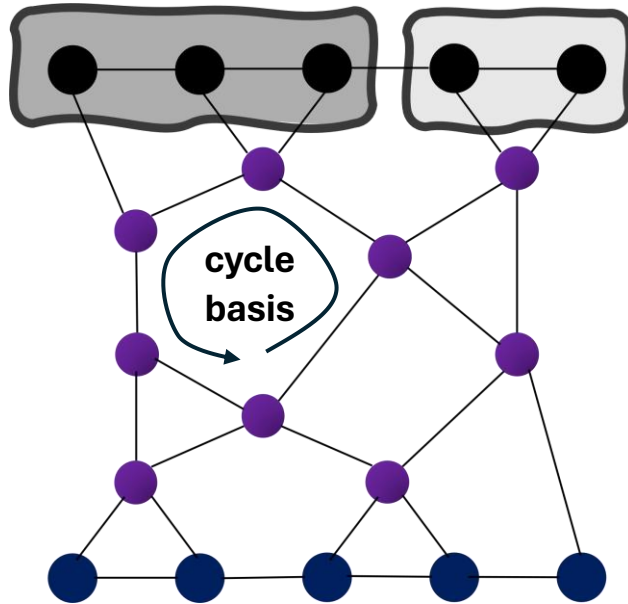
Algorithm for minimal cut

1. The problem of finding the minimal cut is mapped to a graph.



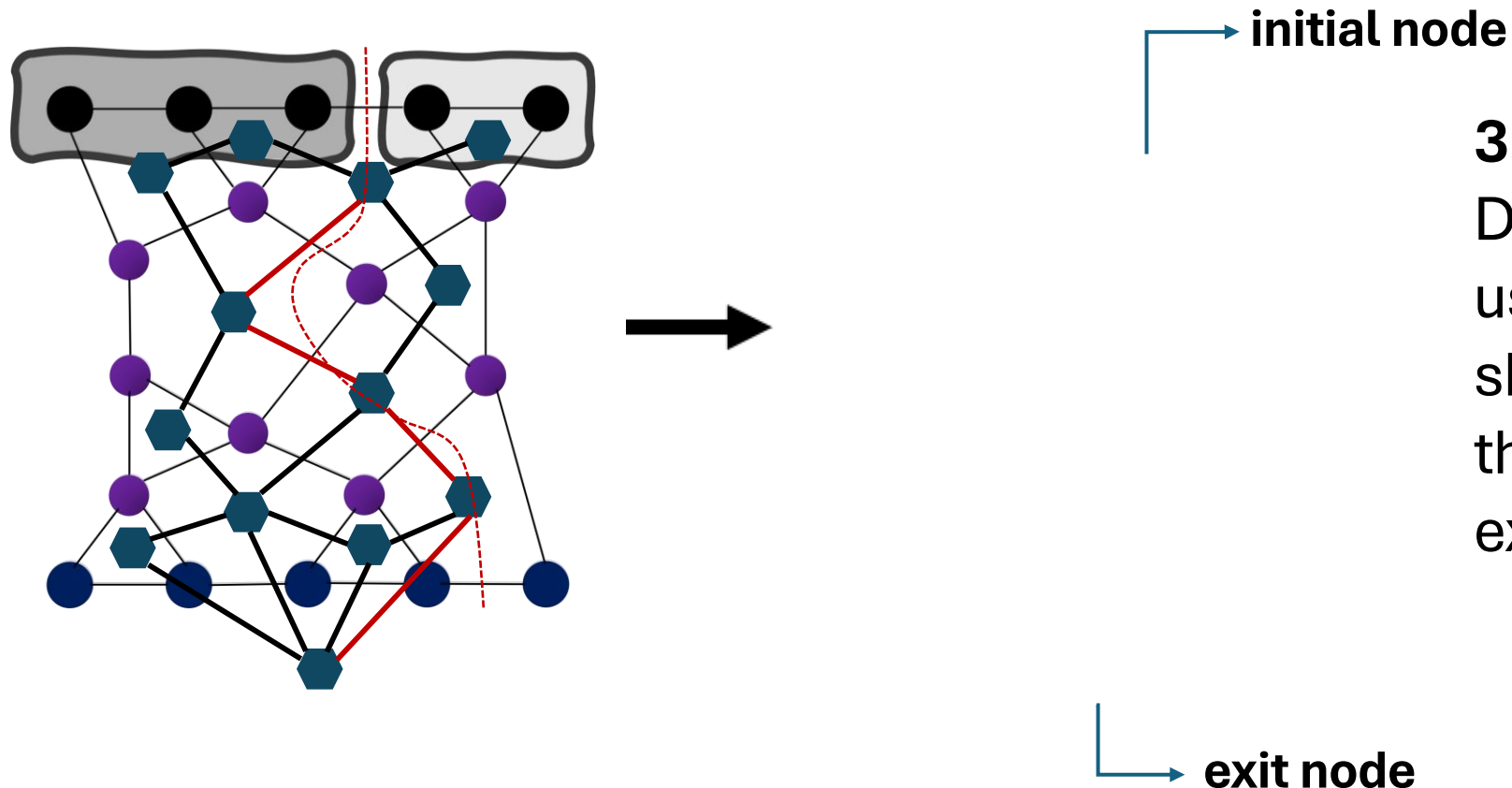
ALGORITHM FOR MINIMAL CUT

2. Another graph is constructed as a representation in cycle basis of the original.



ALGORITHM FOR MINIMAL CUT

2. Another graph is constructed as a representation in cycle basis of the original.

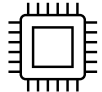


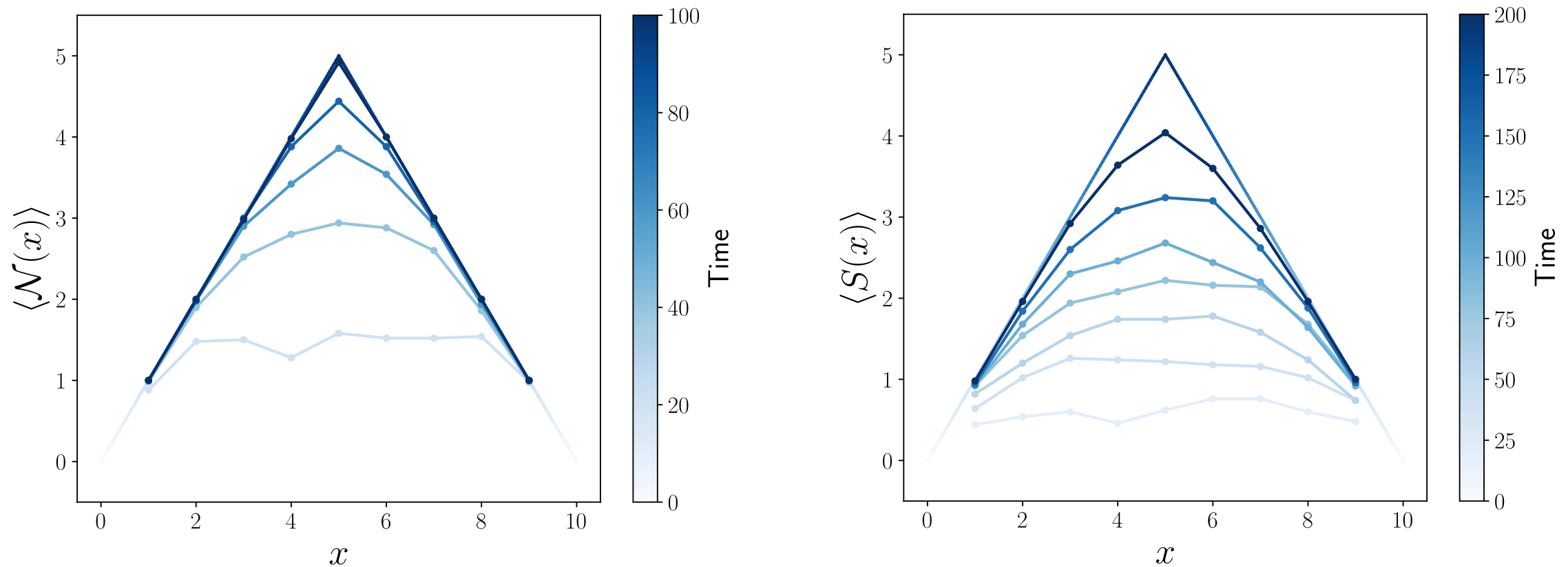
3. In this new graph, Dijkstra's algorithm is used to find the shortest path between the initial node and the exit node.

The total runtime is asymptotically dominated by $\mathcal{O}((L + T)^3)$, where L and T is the number of qubits and gates respectively.

NUMERICAL RESULTS

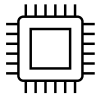
SPATIAL DEPENDENCE OF THE MINIMAL CUT

 In a finite system the asymptotic $\langle S(x, t) \rangle$ profile is that of a pyramid, with a maximum at height $x = L/2$, whose height is $L/2$.

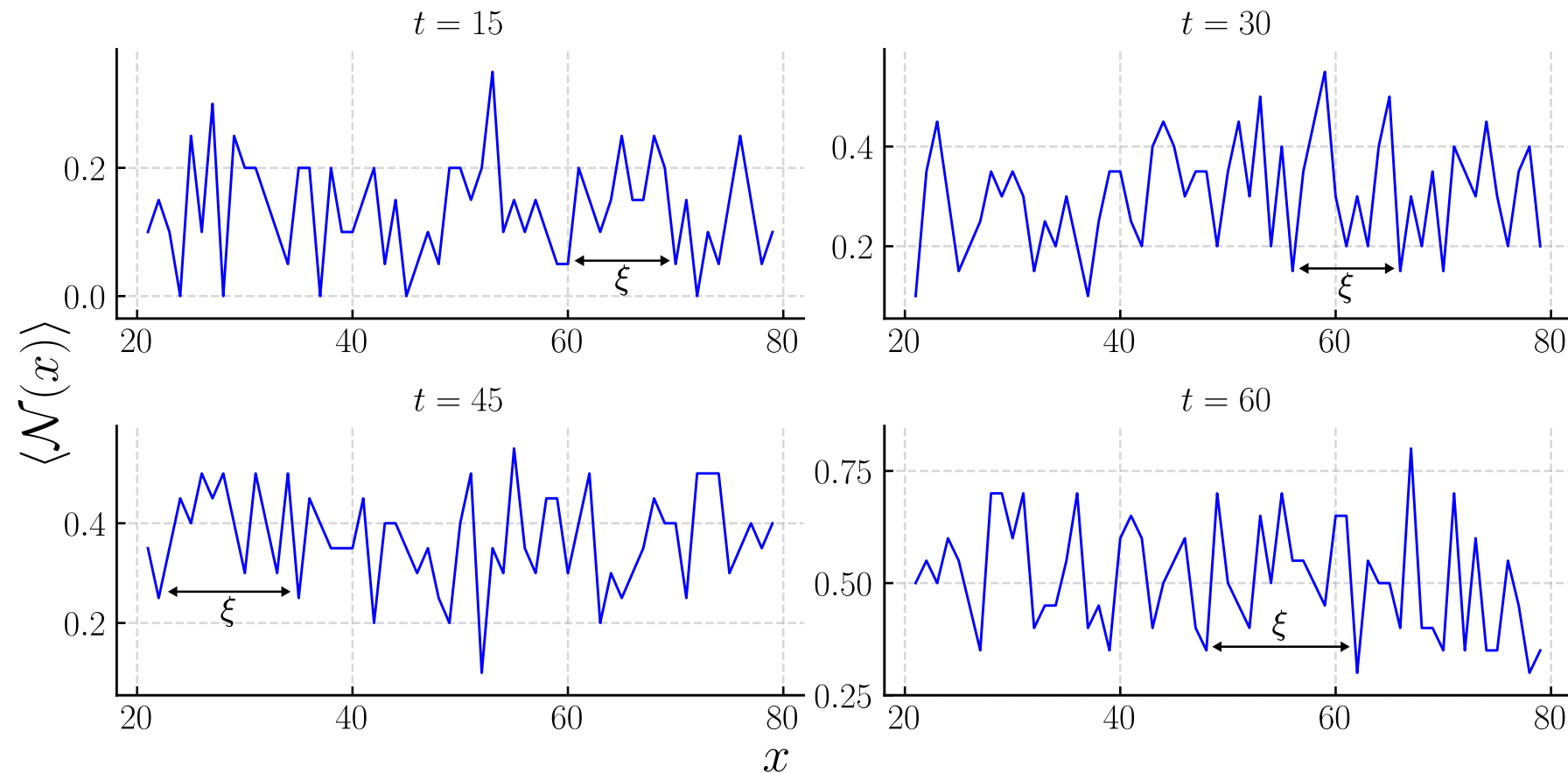


Pyramidal entanglement profile for a chain of $L = 10$ qubits, obtained from an ensemble of 50 random circuits.

SPATIAL DEPENDENCE OF THE MINIMAL CUT



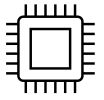
Continuing with the analysis of the spatial dependence of the minimal cut $\mathcal{N}(x, t)$ as a function of the bipartition position x for a fixed circuit depth t .



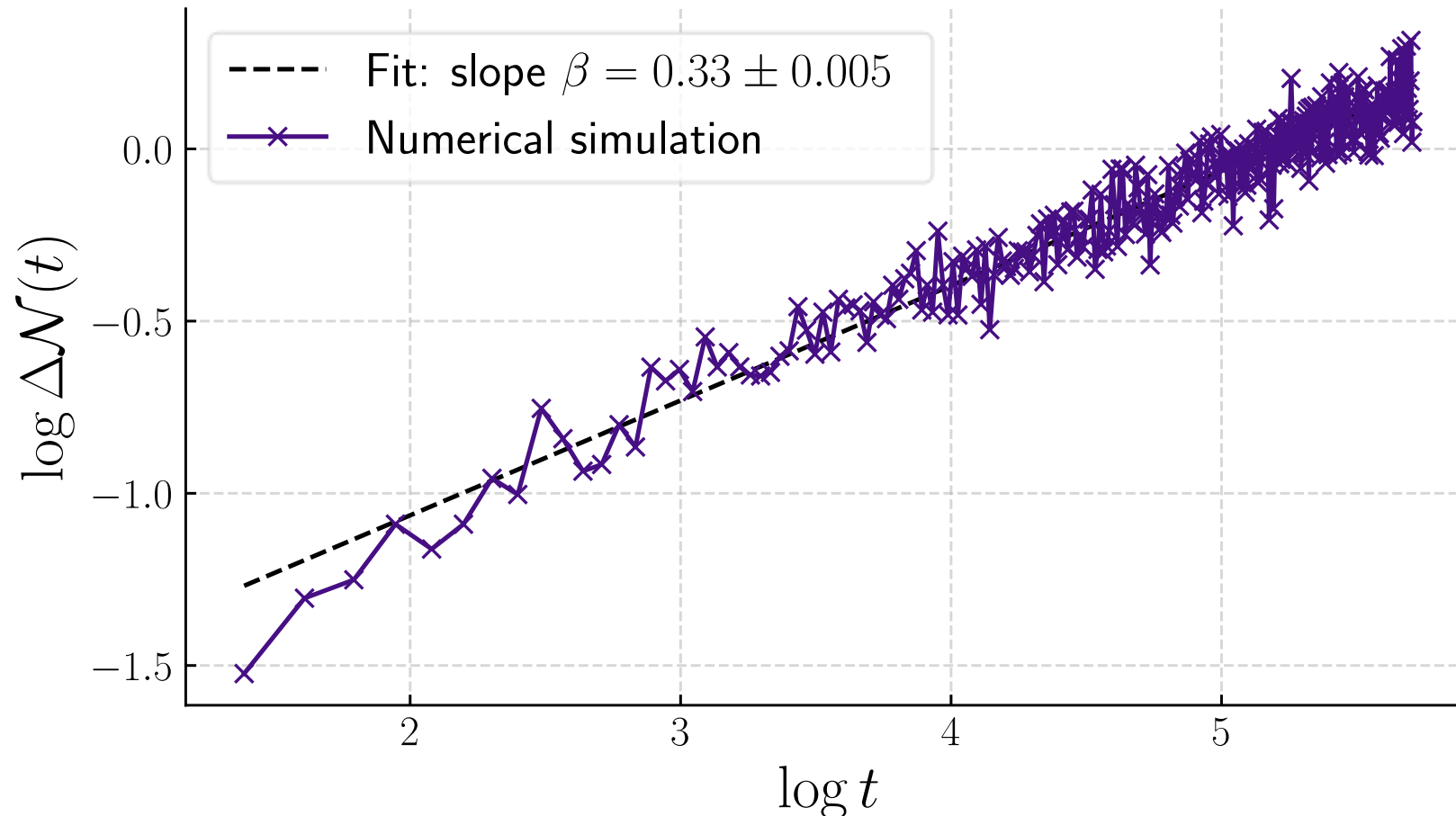
This analysis reveals that the spatial fluctuations scale as $\xi \sim t^{2/3}$, in agreement with KPZ theory

Spatial fluctuations sampled over 100 uniformly distributed random circuits of $L = 100$ qubits at different times $t = 20, 40, 60, 80$.

KPZ FLUCTUATIONS IN MINIMAL CUT



The growth exponent β is extracted from the variance of the minimal cut as a function of time.

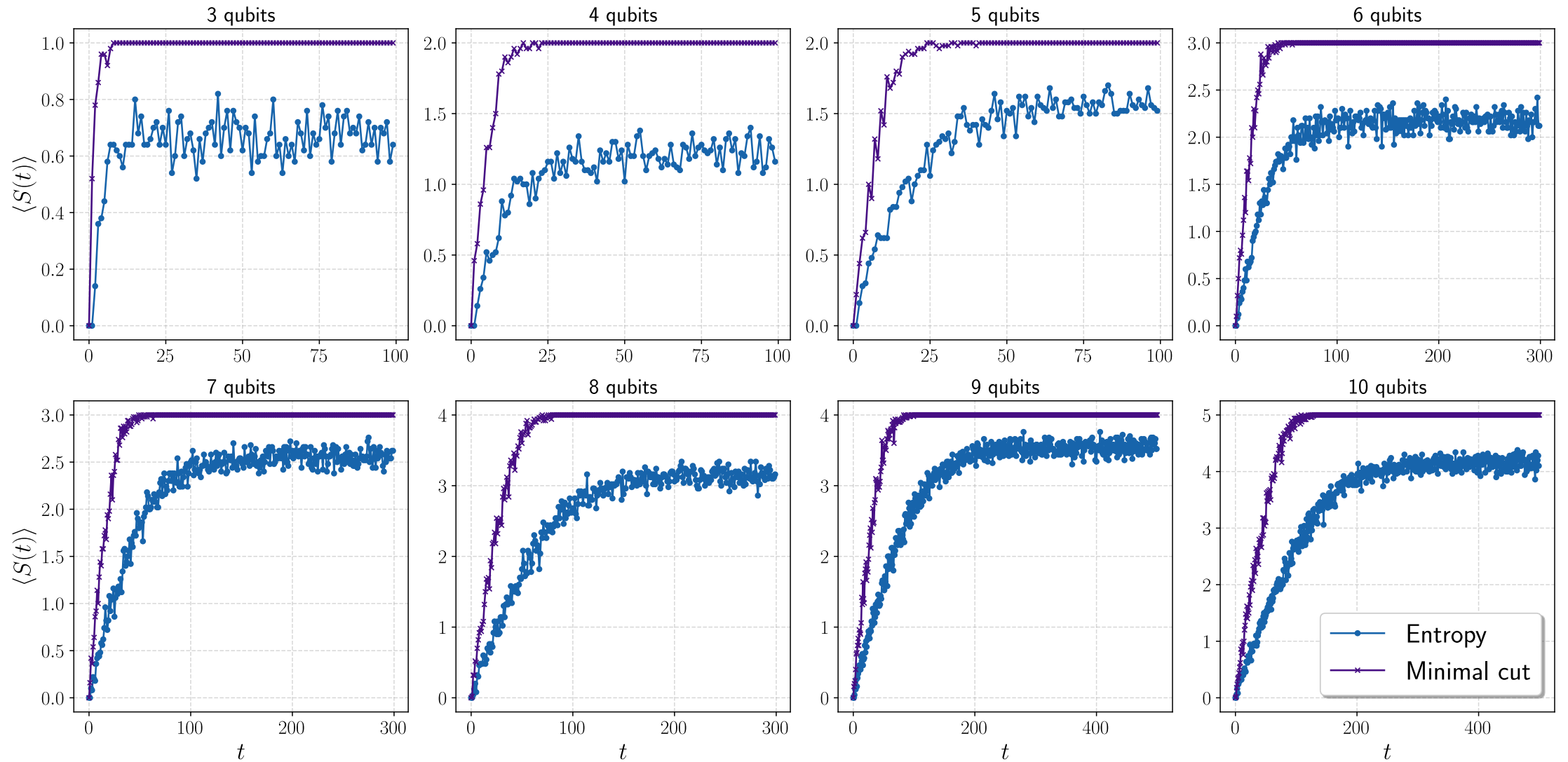


$$\Delta \mathcal{N}(x, t) \sim t^\beta$$

The linear regression in the asymptotic regime yields $\beta = \frac{1}{3}$ in excellent agreement with the KPZ universality class.

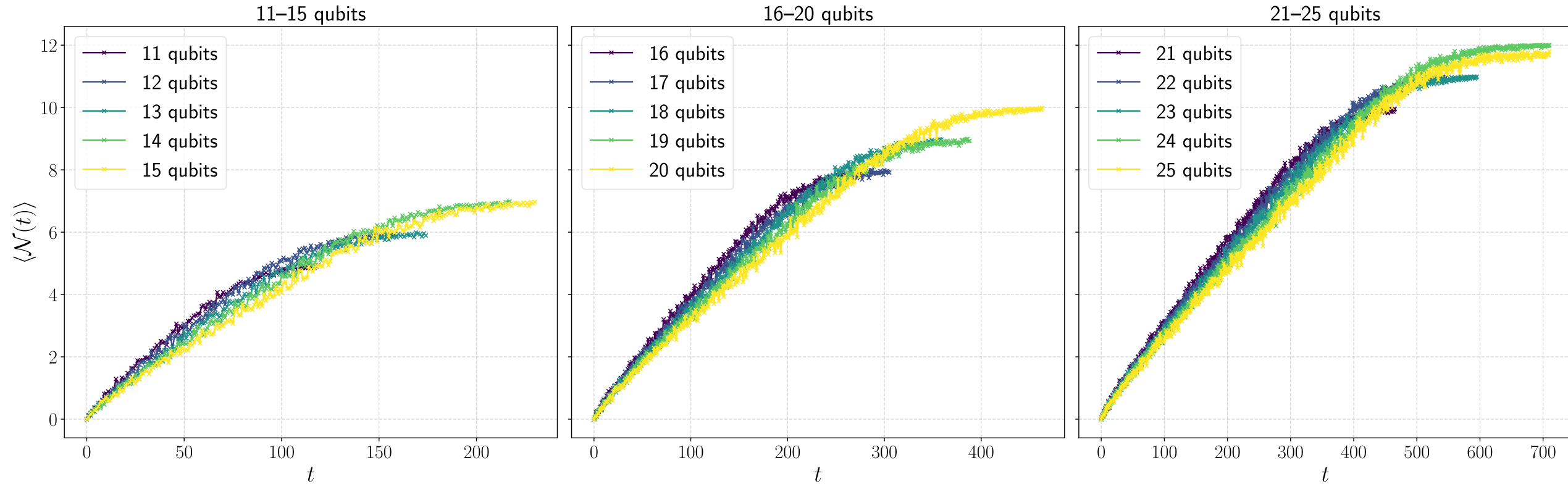
Log-log scaling of the minimal cut fluctuations $\Delta N(t)$ for a random circuit with $L = 50$ qubits, evolved up to depth $t = 300$ and averaged over 100 ensembles.

TEMPORAL DEPENDENCE OF THE MINIMAL CUT



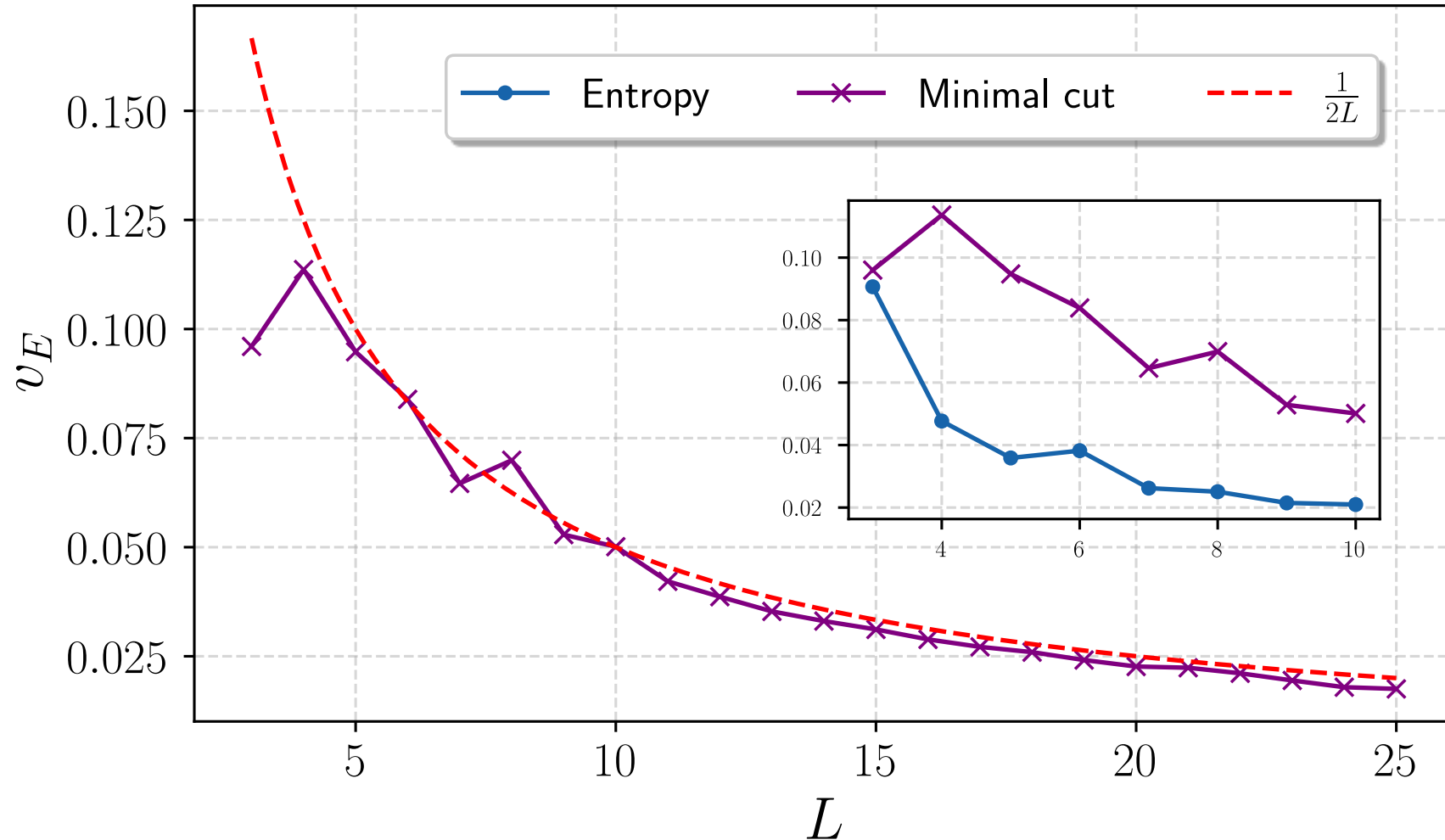
Time evolution of the entanglement entropy and the minimal cut at the central bipartition $x = L/2$, for system sizes ranging from $L = 3$ to $L = 10$ qubits. Each curve corresponds to an average over an ensemble of 100 uniformly distributed random Clifford circuits.

TEMPORAL DEPENDENCE OF THE MINIMAL CUT



Time evolution of the minimal cut for systems with $L = 11$ up to $L = 25$ qubits, considering the bipartition at $x = L/2$, averaged over an ensemble of 100 random circuits. Each system is evolved until reaching its saturation point. A linear regime prior to saturation is observed, with a gradual decrease in the slope as the number of qubits increases.

ENTANGLEMENT VELOCITY



Entanglement velocity v_E extracted from entanglement entropy and minimal cut obtained via linear regression of $S(t)$ and $\mathcal{N}(t)$ versus depth t up to the saturation point.

We find $v_E^{(S)} \leq v_E^{(\mathcal{N})}$, a direct consequence of the inequality $S(t) \leq \mathcal{N}(t)$.

The extracted velocity scales with L as $v_E(L) = 1/2L$, confirming the expectation from the cut growth model.

This implies that the saturation time for a bipartition at position x is,

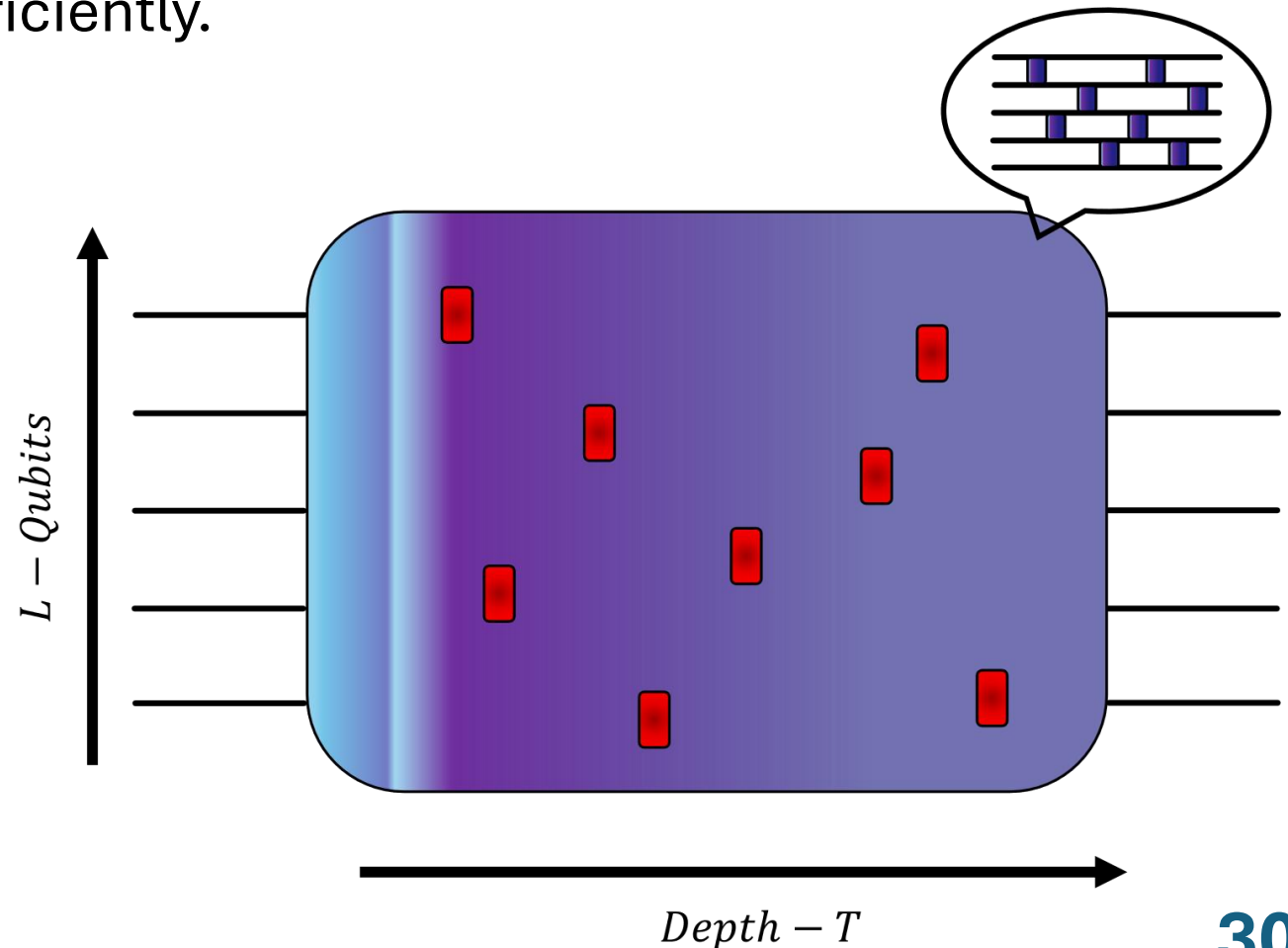
$$t_{sat} = \frac{x}{v_E} = 2Lx.$$

RESEARCH IN
PROGRESS

DOPED CIRCUITS

MOTIVATION

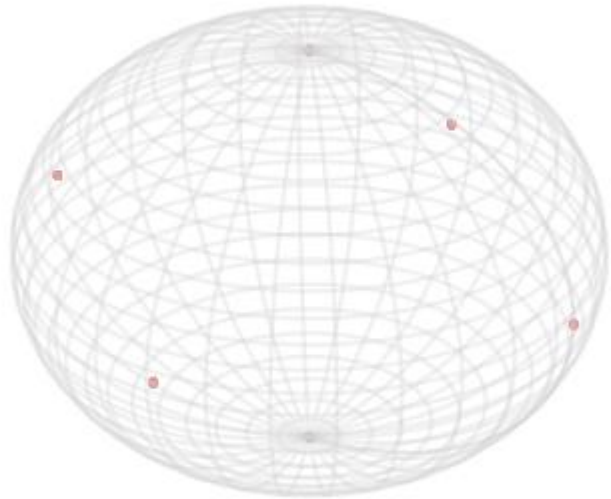
- ❑ Real quantum hardware will likely include not only local two-qubit gates but also higher-size gates (e.g., three-qubit Toffoli), in order to reduce transpilation costs and implement algorithms more efficiently.
- ❑ While the entanglement growth for two-qubit circuits is well understood (mapping to KPZ universality), the behavior with non-local three-qubit gates remains largely unexplored and requires new analysis.



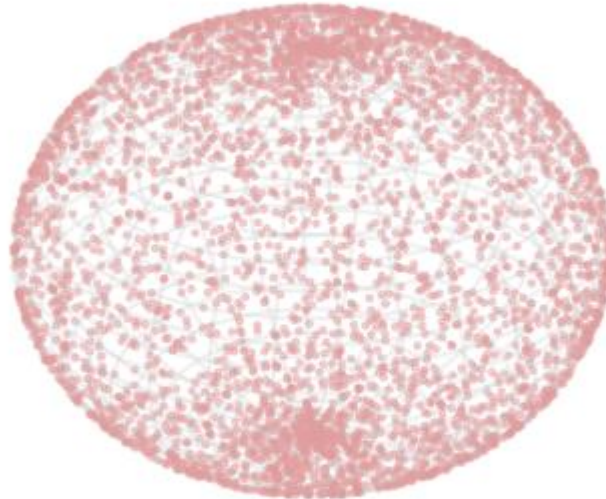
HAAR MEASURE

While Clifford circuits allow efficient classical simulations, they do not capture the genuine statistical properties of quantum entanglement. In contrast, sampling from the Haar measure on the unitary group provides the canonical notion of quantum randomness.

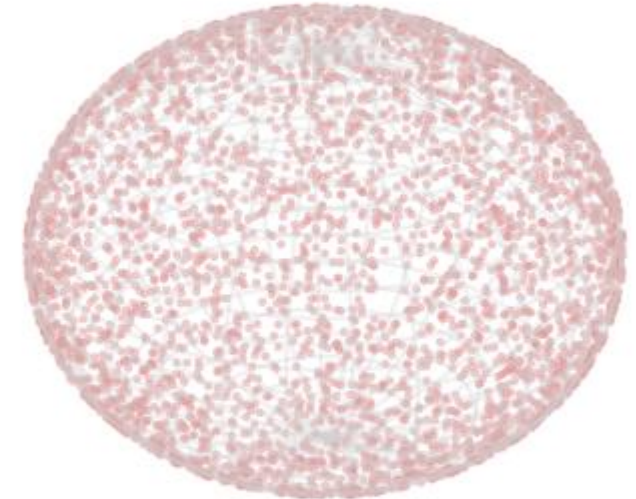
$$d\mu_2 = \sin \theta d\theta \cdot d\omega \cdot d\phi.$$



Clifford Random States

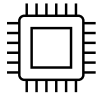


Not Haar (uniform parameters)



Haar random states

LOWER BOUND



The minimal-cut prescription provides an upper bound to the entropy. However, to properly constrain the dynamics, we also need a lower bound, since this gives us a controlled interval within which the entanglement entropy must lie.

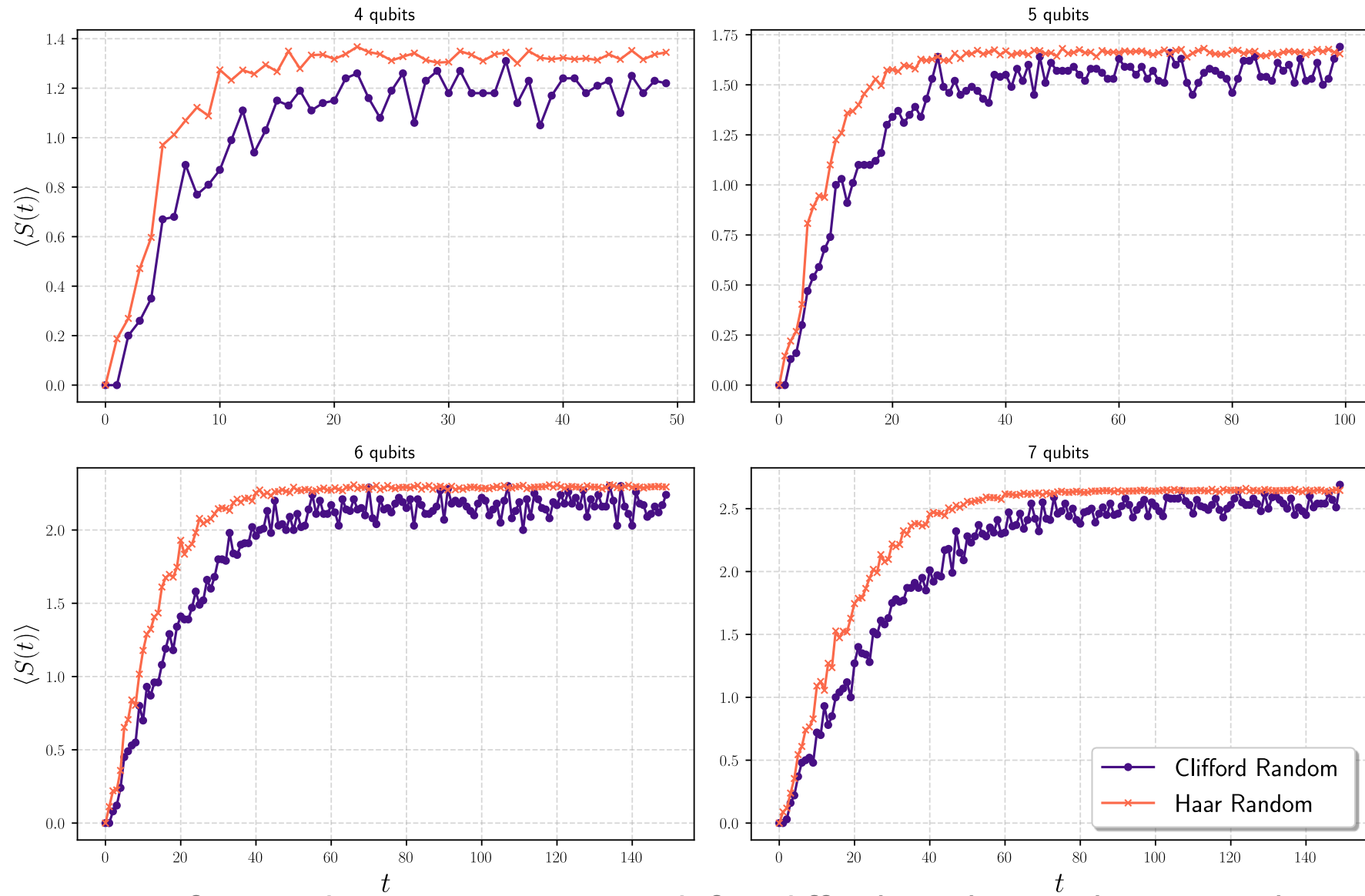
Ballistic lower bound for the entanglement entropy in both doped and undoped circuits. For a bipartition of size $|x|$, the 2nd Rényi entropy is lower bounded by a linear growth law of velocity v :

$$Pr[S_2(x, T) \geq Tv - f(x)] \geq 1 - 2^{-f(x)},$$

where $f(x)$ is any function growing with $|x|$, and the growth velocity is given by,

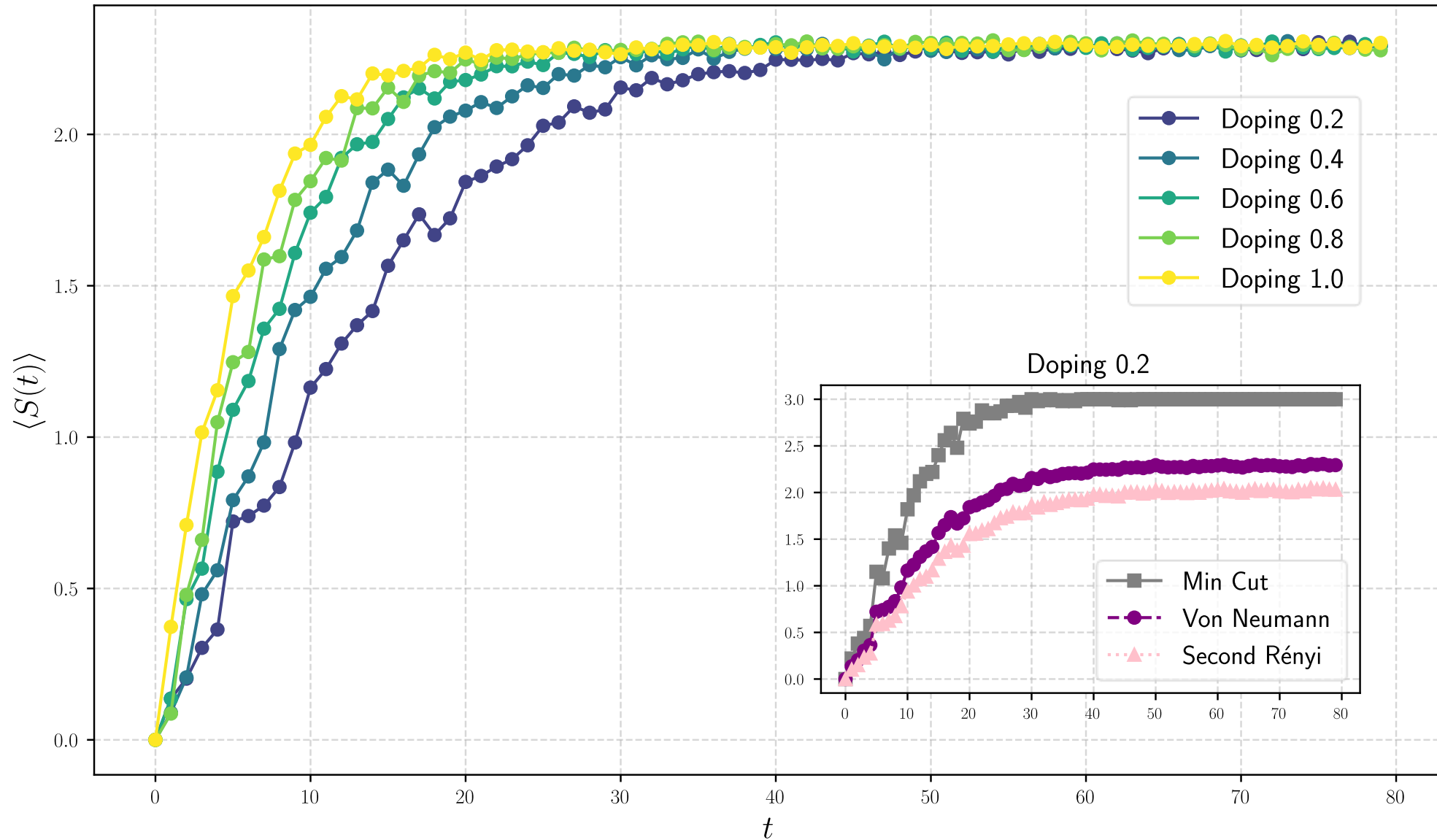
$$v = \left| \log \left((1 - \delta g_{(3)}) \left(1 - \frac{g_{(2)}}{5} \right) + \frac{4}{3} \delta g_{(3)} \right) \right|.$$

PRELIMINAR RESULTS



Comparison of entanglement entropy growth for Clifford-random and Haar-random circuits. Each curve corresponds to an average over 100 ensembles of random circuits with doping parameter $\delta = 0.2$, shown for system sizes of 4–7 qubits.

PRELIMINARY RESULTS

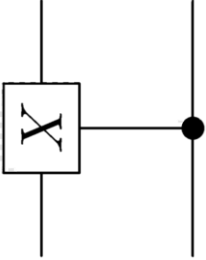
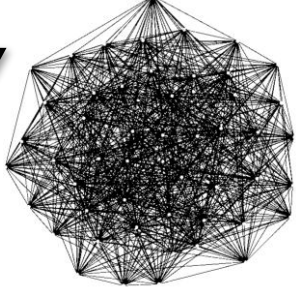


Average entanglement growth for 6-qubit Haar-random circuits at depth 80 over 100 ensembles. The main panel shows von Neumann entropy for different doping values, while the inset compares minimal cut, von Neumann and second Rényi entropies for doping 0.2.

CONCLUSIONS

- ❑ We confirmed that the minimal cut formalism provides a reliable **upper bound** to entanglement growth, reproducing KPZ scaling and revealing the dependence of the entanglement velocity as $v_E = 1/2L$.
- ❑ Preliminary results suggest a transition from Clifford-random to **Haar-random** circuits, paving the way to study genuine quantum randomness in many-body systems, and highlighting the strong interest in further exploring the **doped case** and obtaining tighter information on the **lower bound**.



T  **ANK**   **Y**  **U**